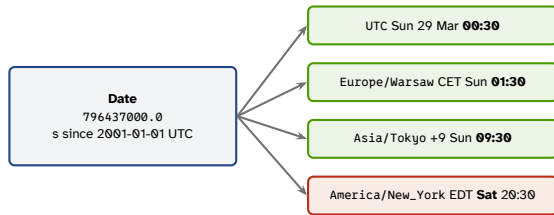


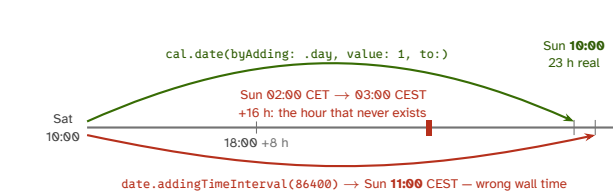
In one line: A **Date** is an **instant**: a Double of seconds since **2001-01-01 00:00 UTC** — no zone, no calendar, no locale. **Calendar + TimeZone + Locale** are the lens that turns it into a human day and does the arithmetic; **formatters** are the (costly) bridge to strings — POSIX/ISO 8601 for machines, localized styles for people.

one instant, four wall clocks



same Date, different day: “stored in UTC, shows the wrong day” = the **display** zone

+86400 s ≠ +1 day (Warsaw, DST 29 Mar 2026)



autumn: a 25 h day, 02:00–03:00 happens *twice*; `startOfDay` ≠ 00:00 where midnight is skipped

How it works

- **Date** — `timeIntervalSinceReferenceDate (2001) / timeIntervalSince1970 (Unix); Date.now (iOS 15)`. Comparing and storing instants needs no zone. JS sends **ms** epochs — divide by 1000.
- **Calendar** carries `timeZone`, `locale`, `firstWeekday`: `date(byAdding:value:to:)`, `dateComponents(_:from:to:)`, `startOfDay(for:)`, `isDateInToday`. `.current` = user settings (maybe Buddhist/Japanese); `pin Calendar(identifier: .gregorian)` for business rules.
- **Days between** = day delta of the two `startOfDays`, never interval / 86400. `DateInterval` for overlap/contains.
- **DateFormatter** — ICU-backed, **expensive to create** (a Time Profiler classic in `cellForRow`) → `static let`. Thread-safe to format/parse (iOS 7+) if no one mutates it. User-facing: `dateStyle/timeStyle` or `setLocalizedDateFormatFromTemplate("MMMd")`, never a raw pattern.
- **Fixed format** (`wire`, `logs`) = `locale` = `en_US_POSIX` + explicit `timeZone`; else a 12-hour or non-Gregorian user setting breaks `HH/yyyy` parsing.
- **ISO8601DateFormatter** — default `.withInternetDateTime`; `.123Z` returns `nil` unless `.withFractionalSeconds (iOS 11)`. `JSONDecoder .iso8601` also rejects fractions → `.custom`.
- **FormatStyle** (iOS 15) — value types, cached by the system: `d.formatted(date: .abbreviated, time: .shortened)`, `.dateTime.weekday(.wide).hour()`, `.iso8601`, `.relative(presentation: .named)`; parse with `Date(str, strategy: .iso8601)`.
- **RelativeDateTimeFormatter** (iOS 13) — “3 h ago”, “yesterday” (`dateTimeStyle` = `.named`); still a formatter: cache it, set calendar.
- **Server vs device** — `wire` = instants (UTC ISO 8601 or epoch); render in the user’s zone. *Local* rules (“9:00 every day”, birthdays, all-day events) = `DateComponents` + a zone **identifier** (Europe/Warsaw), never a fixed offset (+02:00 changes with DST).

Testing — inject time

`Date()` inside logic = untestable + flaky across CI zones. Inject `now: () -> Date` (or a `Clock`, Swift 5.7 / iOS 16) **and** a `Calendar` with a fixed `TimeZone` + `Locale`; test the DST days and 23:59 → 00:00 explicitly. Elapsed time: `ContinuousClock` (counts sleep) / `SuspendingClock` (pauses) or `CACurrentMediaTime()` — **never** `Date()` - `start`: NTP or the user moves the wall clock.

Example

```

enum Formatters {
    // build ONCE, reuse
    static let wire: DateFormatter = {
        let f = DateFormatter()
        f.locale = Locale(identifier: "en_US_POSIX") // fixed fmt
        f.timeZone = TimeZone(identifier: "UTC")
        f.dateFormat = "yyyy-MM-dd'T'HH:mm:ssXXXXX" // yyyy!
        return f }()
}
var cal = Calendar(identifier: .gregorian)
cal.timeZone = TimeZone(identifier: "Europe/Warsaw")!
let next = cal.date(byAdding: .day, value: 1, to: now)!
let days = cal.dateComponents([.day],
    from: cal.startOfDay(for: a), to: cal.startOfDay(for: b)).day
now.formatted(date: .abbreviated, time: .shortened) // UI
let d = try Date("2026-03-29T08:30:00Z", strategy: .iso8601)
  
```

Traps

- ! + 86400 = “24 h later”, not “tomorrow” — 23/25 h days (diagram).
- ! YYYY = *week-based* year: some late-December days print next year. `DD` = day of year, `hh` = 12-hour. Use `yyyy/dd/HH`.
- ! Fixed format without `en_US_POSIX` — works on your phone, fails for a user with 12-hour time or a Buddhist calendar.
- ! `.dateTime` / `dateStyle` output on the wire — localized, unparsable.
- ! Mutating `dateFormat/locale` of a shared formatter while another thread formats — configure once, then read-only.
- ! Off-by-one “expires in N days” near midnight — compared instants, not day boundaries in the *user’s* zone.
- ! Client-side time gates (trial, rate limit) — the user can roll the clock back; enforce on the server.

Pattern letters (Unicode TR35)

yyyy	calendar year	YYYY	week-based year
MM · MMM	03 · Mar	dd · DD	day of month · of year
HH · hh	0–23 · 1–12	a	AM/PM
mm · ss	minute · second	SSS	fraction
E · EEEE	Sun · Sunday	XXXX	Z or +02:00

Remember

Instant in, instant out; the zone is chosen at the edge. Add *days* with a `Calendar`, *durations* with a `Clock`, and never write `YYYY`.